

Verifier-Assisted versus One-Shot LLM Intent→Configuration: A Paired NetOpsTraceBench Study

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CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed NetOpsTraceBench, a paired comparison of a one-shot LLM intent→configuration pipeline and a verifier-assisted generate-validate-repair pipeline on 1,200 intent→config episodes. Using gpt-5-mini, a deterministic NetOps validator, and Mininet-style emulation, we applied a preregistered binary episode success rule (validator accepts final config AND emulator shows no availability degradation above tolerance AND no automatic rollback). Of 1,200 planned paired tasks we completed 1,199 pairs. Baseline success = 1,196/1,199 (0.9974979149); guarded success = 1,199/1,199 (1.0000000000). The paired-success-rate difference (guarded - baseline) = 0.0025020851 (95% bootstrap percentile CI [0.0000, 0.0058381985], bootstrap_seed=1729, resamples=10,000); exact McNemar two-sided $p = 0.25$. The preregistered power target sought a 5 percentage-point minimum detectable effect; given the observed near-ceiling baseline, the study lacked power to detect much smaller absolute gains. Representative validator traces and provider audit logs show repair was invoked only three times. Under these controlled emulation and task-corpus conditions we find no statistically significant advantage for the verifier-assisted pipeline; we report artifacts, analytic deviation (Wilcoxon → exact McNemar for binary outcomes), and detailed execution accounting to support reproducible interpretation.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent expressed in natural language into executable device configurations for network operations (NetOps). Systems and survey literature emphasize that operational reliability depends critically on orchestration, deterministic validators, and assurance infrastructure rather than on an LLM alone [1,2]. Generate-validate-repair (verifier-assisted) architectures are a canonical mitigation strategy: an LLM proposes candidate outputs, an automated verifier checks constraints and safety, and an automated repair loop iterates until the candidate satisfies verifier rules [3–5]. Despite conceptual support, publicly archived, preregistered, paired evidence measuring whether verifier assistance yields an operationally meaningful improvement (rollback-aware binary success) relative to one-shot generation is sparse.

We address this gap with NetOpsTraceBench: a preregistered, paired, emulator-grounded study comparing (A) baseline one-shot NL→config generation and (B) a guarded generate-validate-repair pipeline using the same shared initial candidate. The study asked: Can a reproducible, workflow-centered benchmark of paired intent→config episodes produce an objective paired binary success metric that reliably distinguishes verifier-assisted pipelines from one-shot generation across two

simulated topologies? Our contributions are (1) a preregistered experimental design and full execution artifacts (study ID rX4f7-1); (2) a paired analysis on 1,199 completed pairs using gpt-5-mini and a deterministic validator; (3) transparent reporting of an analytic deviation (final confirmatory test used an exact McNemar test appropriate for paired binary outcomes) and execution accounting that explains why the guarded pipeline rarely required repair.

II. RELATED WORK

Recent survey and system literature frames LLMs as components inside broader AIOps/NetOps workflows and repeatedly emphasizes that correctness and trustworthiness are properties of the whole orchestration stack (model + verifier + provenance + sandboxed execution), not of the base LLM alone [1–3]. Multiple community reviews urge workflow-centered evaluation—trace quality, validator interactions, and rollback-aware metrics—rather than isolated one-shot QA tests; these reviews motivate reproducible, audit-ready benchmarks that combine generation, deterministic validation, and emulation [1,2,4].

Concrete verifier-assisted approaches appear across adjacent domains and NetOps prototypes. Work on NL→formal translations and CTL/specification generation demonstrates generator+verifier loops where formal checks drive iterative repairs; similarly, topology-compilation systems translate high-level security intent into executable topologies with compliance checks and constraint enforcement [5,6]. Prototype efforts in the configuration space (text→config generators and misconfiguration detectors) and domain-specific "LLM firewalls" illustrate practical patterns: an LLM proposes a candidate, an automated validator enforces syntactic and semantic constraints, and an orchestrator optionally triggers constrained repair or rejects the proposal [7–10]. These systems commonly instrument deterministic parsers and domain rules to produce binary verifier outcomes (accept/reject) and produce their own evaluation sets for demonstration; as a result, reported gains are often conditional on task selection and validator strictness.

Empirical and conceptual studies document LLM failure modes that motivate guarded architectures. Recent analyses identify hallucination, unit/semantic errors, overconfidence/miscalibration, provenance loss, and susceptibility to prompt-injection as recurrent risks when LLMs propose executable artifacts—risks that an external verifier or prove-

nance linker can detect or mitigate in some instances but not eliminate universally [11–14]. Several papers demonstrate that verifier-guided repair reduces syntactic and some semantic errors in constrained settings, but they also highlight sensitivity to the verifier’s rule set (strictness) and to the representativeness of evaluation examples. In short, prior work shows feasibility of generate–validate–repair patterns but also underscores the dependence of measured improvement on corpus difficulty, adversarial content, and verifier configuration.

Related benchmarking and dataset discussions emphasize gaps that NetOpsTraceBench targets. System papers often construct private or narrowly scoped held-out sets calibrated to their prototype; TopoIntent and other recent tools note the absence of large public, vendor-diverse intent→config datasets with execution ground truth and rollback semantics, which complicates cross-system comparisons [6,7,9]. Surveys therefore call for standardized, workflow-centered benchmarks that (a) record full provider and validator traces, (b) embed emulator execution to check operational side effects (availability/rollback), and (c) preregister analysis choices to reduce researcher degrees of freedom [1,2]. NetOpsTraceBench operationalizes these recommendations by preregistering inclusion/exclusion rules, seeds, a binary, rollback-aware episode success metric, and by archiving raw model outputs, validator traces, and execution manifests so that the effect of verifier design and task corpus can be examined reproducibly.

Taken together, the literature supports the conceptual value of verifier scaffolding while also documenting two recurring caveats that our study made explicit in design and interpretation: (i) measured gains depend strongly on task difficulty and verifier thresholds, and (ii) many demonstrations use constructed datasets and domain-specific validators, which limits generality. NetOpsTraceBench contributes a preregistered, artifact-rich paired evaluation that makes these dependencies observable (repair invocation counts, contingency discordances, and emulator rollback checks) rather than inferred, enabling follow-up sensitivity studies that vary corpus adversariality and verifier strictness.

III. METHODOLOGY

Overview and preregistration: The study was preregistered as NetOpsTraceBench (study_id = rX4f7-1) and the preregistration manifest (SHA-256 = 77abe8b493bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177a7ed5712) froze the experimental plan before any model calls. The preregistration enumerated confirmatory hypotheses (H1–H3), inclusion/exclusion rules, contamination thresholds, seeds, stopping rules, analysis executor (paired_binary_analysis_v1), primary estimand name (paired_success_rate_difference_guarded_minus_baseline), and the prespecified power target (MDE = 5 percentage points; $\alpha=0.05$; power 0.8) given assumed baseline ≈ 0.70 and discordant-pair assumptions recorded in the plan.

Task corpus and topology clusters: The benchmark scope defined 1,200 distinct operator-intent tasks split into two simulated topology clusters: edge (600) and core (600).

Each task is generated deterministically by deterministic_netops_task_generator_v5 (generator_version = 5.0) using frozen per-task generation_seed values. Task payloads contain structured fields: initial_state, expected_final_state, explicit_required_sequence arrays, allowed_touched_fields, workflow_policies, and a declared difficulty level (e.g., "medium", "hard", "very_hard"). Tasks are labeled synthetic or consented and carry deterministic provenance metadata used by contamination checks. Task and schema versions are 1.0.

Contamination and inclusion/exclusion: The preregistration specified conservative contamination detection: token/subsequence overlap threshold $T_{\text{contam}} = 0.85$ (normalized overlap) and optional embedding-similarity check (cosine threshold 0.92) for flagged items. Pairs were excluded automatically if contamination checks exceed thresholds, if both episodes fail to produce parsable vendor-rendered configs after one automated re-execution attempt, or if emulator instrumentation deterministically fails to measure availability. Exclusions are deterministic, logged, and reported; no human adjudication occurs. The missingness plan defines primary analysis as complete_pair_primary (exclude flagged pairs) and a sensitivity analysis that treats excluded pairs as failures (complete_pair_primary_with_failure_as_zero_sensitivity).

Pipeline implementations and orchestration semantics: We implemented two paired conditions per task using a shared initial candidate generation approach (generation_semantics = "shared_initial_candidate"). Adapter orchestration used hosted_netops_gvr_v1. Model requests used the requested_model = gpt-5-mini via provider recorded as openai_responses; execution/model_configuration.json records requested_model = gpt-5-mini, max_output_tokens = 1500, maximum_attempts_per_call = 1. Per preregistration, initial_generation_calls_per_task = 1 and maximum_repair_calls_per_task = 1; retrieval_augmented_generation = false; independent_condition_generation = false (guarded and baseline share the same initial candidate in each pair to isolate verifier and repair effects).

Baseline (one-shot) condition: For each task the LLM receives the frozen, schema-specified intent and returns a single candidate configuration (the shared_initial_candidate). The baseline episode records that candidate and proceeds to deterministic parsing and emulator preflight checks without invoking the automated repair loop.

Guarded (verifier-assisted) condition: The guarded pipeline feeds the same shared_initial_candidate to deterministic_netops_validator_v5 (validator_version = 5.0). If the validator accepts the candidate, the episode proceeds to emulator execution; if the validator rejects the candidate the pipeline may invoke an automated repair iteration (a single additional LLM call per preregistered maximum_repair_calls_per_task = 1) that produces a repaired candidate which is then revalidated. Repair invocation is conditional and limited by the frozen retry policy; all repair prompts, provider traces, and repair results are archived. The scoring rule used by the validator is recorded as "full_intent_constraint_validation" and scor-

ing_status entries are emitted (COMPLETED or error codes) into per-episode scoring artifacts.

Deterministic validator and emulator checks: deterministic_netops_validator_v5 performs canonical parsing, counts parsed_command_count and required_command_count, enforces workflow_policies (e.g., must_be_down_for_fields, minimum_active_in_groups), and produces a normalized_configuration string. The preregistered binary episode success requires (1) validator acceptance of the final configuration (validation_after.valid == true), (2) emulator execution showing no availability degradation above the preregistered tolerance, and (3) no automatic rollback event during simulated execution. Validator traces include structured diagnostics: observed_touched_field_count, violation_codes, difficulty metadata (changed_interface_count, dependency_rule_count, pattern), and repair_applied boolean. These diagnostics were used for exploratory failure-mode catalogs.

Analysis plan and confirmatory procedure: The preregistration specified a paired analysis using paired_binary_analysis_v1. The primary estimand is the paired_success_rate_difference_guarded_minus_baseline. The preregistration anticipated using a nonparametric paired test and bootstrap CI computation. Because observed outcomes are binary, the archived confirmatory analysis executed an exact McNemar two-sided test for paired binary outcomes and computed a paired bootstrap percentile 95% CI using frozen_bootstrap_seed = 1729 and bootstrap_resamples = 10,000; both the contingency table and bootstrap procedure are archived. Missingness, exclusions, and deviations are reported deterministically; no post-hoc inclusion decisions were made.

Stopping rules, provider audit, and execution accountability: The run obeyed the preregistered stopping rule (halt if model calls reach maximum_model_calls = 2,400). Execution produced provider_call_audit entries: hosted_model_call_event_count = 1,203; completed_hosted_model_call_event_count = 1,202; failed_hosted_model_call_event_count = 1; stage_counts: shared_initial_generation = 1,200 and repair = 3. Execution manifest and deterministic_reconciliation artifacts track completed_episode_count = 2,398, failed_episode_count = 2, and complete_pair_count = 1,199. All raw_model_outputs, provider traces, task manifests, transformation manifests, validator traces, scoring artifacts, and analysis logs (including analysis/paired_contingency_table.csv, analysis/condition_summary.csv, analysis/missingness_summary.csv, and analysis/paired_difference_figure.svg) are archived to permit exact replay of the analysis executor and CI computation.

Power rationale and sensitivity: The preregistered power plan targeted an absolute paired increase of 5 pp favoring guarded (MDE = 0.05) under assumed baseline ≈ 0.70 and discordant-pair assumptions; those assumptions and sensitivity grids (p_0 in [0.5,0.8], discordant rates in [0.1,0.4]) are stored in the preregistration. The manifest records that the observed

baseline was near ceiling (≈ 0.9975), which reduces power to detect small absolute improvements; the archived bootstrap CI and discordant counts provide the quantitative basis for that interpretation.

Reproducibility and artifact governance: All seeds, code for task generation, validator rules, repair prompts, provider traces, and analysis code are published in the run artifacts. The run produced artifact_count = 8,408 and a full hash manifest; representative per-episode scoring JSONs (schema version 1.0) show the normalized_response, validator_trace entries, and score_reason_code values used to compute the binary success outcomes. The methodology deliberately prioritizes deterministic components and explicit logging to ensure that the paired comparison isolates the verifier/repair mechanism while keeping generation variability constrained by shared_initial_candidate semantics.

IV. RESULTS

Execution and accounting (archived): Planned episode_count = 2,400 (1,200 pairs); completed_episode_count = 2,398; failed_episode_count = 2; completed_pairs = 1,199. Provider audit: hosted_model_call_event_count = 1,203; completed_hosted_model_call_event_count = 1,202; failed_hosted_model_call_event_count = 1. Stage counts show shared_initial_generation = 1,200 and repair = 3 (i.e., the guarded pipeline triggered repair calls only three times). Artifact_count = 8,408 and the run completed at 2026-08-30T03:15:12.141890+00:00; representative artifact examples and validator traces are archived (e.g., execution/scoring/task-000001-baseline.json).

Primary confirmatory outcomes (archived analysis): complete_pair_count = 1,199; baseline_success_count = 1,196 (baseline_success_rate = 0.9974979149291076); guarded_success_count = 1,199 (guarded_success_rate = 1.0). Paired contingency: n11 = 1,196; n10 = 3 (guarded=1 baseline=0); n01 = 0; n00 = 0 (see analysis/paired_contingency_table.csv). Paired-success-rate difference (guarded - baseline) = 0.002502085070892446. Exact McNemar two-sided $p = 0.25$. Paired bootstrap percentile 95% CI (bootstrap_seed = 1729; 10,000 resamples) = [0.0000, 0.005838198498748957]. Missingness summary: failed_baseline_episodes = 1; failed_guarded_episodes = 1; both_missing_pairs = 1 (see analysis/missingness_summary.csv).

Representative diagnostics: Validator traces in scoring artifacts show that, for typical tasks, the initial shared candidate satisfied intent constraints and validator checks (e.g., execution/scoring/task-000001-baseline.json shows validation_after.valid = true and repair_applied = false). The low repair invocation count (3) is corroborated by the provider audit stage_counts. These artifact-grounded diagnostics indicate that under the chosen validator configuration and synthetic task corpus the LLM initial candidates were accepted in the vast majority of episodes, leaving little headroom for guard-triggered improvement.

V. DISCUSSION

Statistical and operational interpretation: The exact McNemar test and bootstrap CI both indicate no statistically significant advantage for the guarded pipeline under this corpus and validator configuration ($p = 0.25$; 95% CI includes zero). Two operational explanations are supported by archived artifacts. First, near-ceiling baseline performance ($\approx 99.75\%$) constrained absolute headroom for improvement; the pre-registered MDE was 5 percentage points, so the run was not powered to detect the observed ≈ 0.25 percentage-point effect. The archived CI and discordant counts ($n_{10} = 3$, $n_{01} = 0$) quantify this limitation and show that detection of much smaller practical gains would require substantially more discordant pairs or a task corpus with lower baseline success.

Second, corpus and verifier design materially affected repair invocation. Representative task_manifest entries and scoring traces show tasks were synthetic with explicit required sequences and conservative initial generators (deterministic_netops_task_generator_v5), and the deterministic validator (deterministic_netops_validator_v5) accepted the majority of initial candidates. Those design choices—synthetic tasks yielding clear required sequences and a validator configuration that enforces the preregistered intent constraints—likely reduced baseline difficulty and therefore the frequency of repairs. The provider audit (repair = 3) and scoring artifacts (validation_after.valid = true for typical episodes) jointly support this interpretation.

Threats to validity and externality: Internal validity is strong for deterministic components (frozen seeds, deterministic task generation, and automated exclusion rules), but construct validity depends on whether the binary success rule (validator acceptance + emulator availability tolerance + no rollback) maps to operationally meaningful correctness across production environments. External validity is limited: only one requested model family (gpt-5-mini) was exercised, task examples were synthetic/consented and vendor-agnostic, and emulation (Mininet-style) may not capture vendor idiosyncrasies and routing convergence phenomena. Conclusion validity: the very low discordant counts imply limited power to detect small absolute differences; the preregistered MDE (5 pp) should be considered when interpreting the null result.

Operational implications: When initial generation already meets deterministic verifier criteria for common, well-specified tasks, generate-validate-repair will be infrequently exercised and measured gains on a binary success metric will be small. To evaluate verifier value, future benchmarks should increase task ambiguity/adversarial content, vary verifier strictness, and exercise multiple model families. The archived artifacts and traces provide a reproducible starting point to perform such sensitivity studies.

VI. LIMITATIONS

- Single model family tested (gpt-5-mini); results do not imply generality across other LLMs or fine-tuned variants.

- Task corpus skew: synthetic/consented tasks yielded near-ceiling baseline success ($\approx 99.75\%$), limiting headroom and statistical power to detect practical improvements by guarded pipelines.
- Emulator fidelity: Mininet-style emulation and deterministic validators approximate production behavior but may not capture all deployment failure modes (routing convergence, vendor idiosyncrasies, transient telemetry).
- Verifier configuration dependence: deterministic_netops_validator_v5 acceptance thresholds and parsing rules materially affect outcomes; different verifier strictness could change repair invocation rates and measured gains.
- Analytic deviation documented: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived confirmatory analysis used an exact McNemar test because outcomes were binary. This deviation is recorded in the archived analysis artifacts and in the Disclosure Statement above.
- Operational external validity: absence of heterogeneous vendor renderers, real production templates, and live rollouts constrain direct production generalization.

VII. CONCLUSION

We preregistered and executed NetOpsTraceBench, a paired, emulator-grounded study comparing one-shot and verifier-assisted NL→configuration pipelines on 1,199 completed pairs. Under our synthetic task corpus, deterministic validator settings, and gpt-5-mini as requested, baseline one-shot generation already achieved near-ceiling success ($\approx 99.75\%$). The guarded pipeline increased absolute success by ≈ 0.25 percentage points (paired difference = 0.0025020851) with exact McNemar two-sided $p = 0.25$ and a 95% bootstrap CI that includes zero. Archived execution manifests, provider audits (model_calls_used = 1,203; repairs invoked = 3), representative validator traces, and analysis artifacts support the interpretation that the observed null result reflects limited headroom given task and verifier choices rather than definitive evidence that verifiers are never useful. Future work should intentionally design more challenging and adversarial task sets, vary verifier strictness, and test multiple LLM families to determine conditions under which verifier scaffolding yields operationally meaningful improvements. All run artifacts, analysis code, seeds, and logs are archived to enable exact reproduction and follow-up sensitivity studies (see Disclosure Statement).

DISCLOSURE STATEMENT

Mandatory disclosure (preregistered run rX4f7-1). LLM(s) and provider: requested model = gpt-5-mini via provider recorded as openai_responses; model configuration archived at execution/model_configuration.json (requested_model=gpt-5-mini, max_output_tokens=1500). Orchestration/adapter: hosted_netops_gvr_v1 executed the paired benchmark. Deterministic tooling: deterministic_netops_validator_v5 performed canonical

parsing, intent-constraint validation, and emulator preflight checks; `deterministic_netops_task_generator_v5` produced synthetic tasks. Exact initial master prompt: archived at `provenance/master_prompt.txt` (immutable reference SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Topic constraint and autonomy boundary: the human Research Director selected the topic family and approved the master prompt pre-lock; after the autonomy lock the AI autonomously formulated the research question, conducted literature review, generated the preregistration, executed the experiment, performed analysis, and wrote and revised the manuscript. Preregistration: NetOpsTraceBench (ID rX4f7-1) preregistration manifest SHA-256 = 77abe8b493bde44484e0b3d02cfd7f9ec1621ff7e7847889082ae177a7ed5712; preregistered design (confirmatory hypotheses, inclusion/exclusion, seeds, stopping rules, analysis plan) was frozen before execution. Execution summary: started 2026-08-30T01:21:55.415955+00:00, completed 2026-08-30T03:15:12.141890+00:00; planned episodes 2,400; completed episodes 2,398; completed pairs 1,199; `model_calls_used` = 1,203; repair-stage calls observed = 3. Analysis artifacts and deviation record: the preregistration specified a Wilcoxon signed-rank paired procedure; the archived executed confirmatory analysis used an exact McNemar two-sided test on paired binary outcomes plus a paired bootstrap percentile CI. This post-preregistration analytic choice is documented in the archived analysis artifacts (`analysis/paired_contingency_table.csv`, `analysis/results.json`, `analysis/analysis_log.jsonl`, `analysis/paired_difference_figure.svg`) produced as part of the run that completed at 2026-08-30T03:15:12.141890+00:00. Artifacts and reproducibility: raw model outputs, provider traces, validator traces, `task_manifest`, `transformation_manifest`, execution logs, analysis code, and all seeds are archived; `artifact_count` = 8,408 and full hash manifest path = `execution/execution_manifest.json`. Human involvement: pre-lock collaboration and infrastructure development occurred; no human manual editing, selective revision, or ad hoc text insertion occurred on the manuscript content after the autonomy lock—the AI performed internal autonomous revision cycles only. Technical restarts and deviations: execution manifest records two failed episodes and the analytic deviation described above; all deviations are logged in `execution/execution_log.jsonl`. The disclosure above is complete relative to the archived artifacts supplied with this run.

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